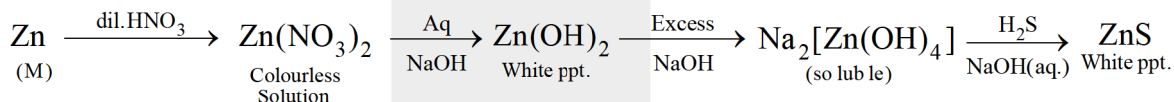
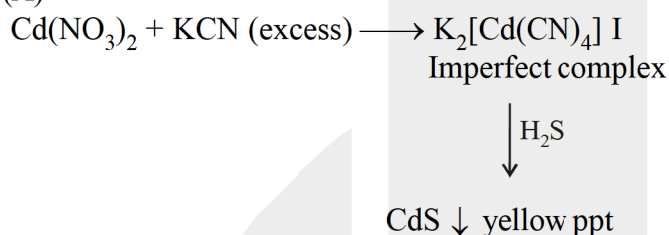


HINTS & SOLUTIONS

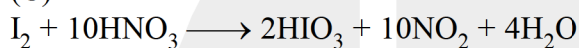
Q.1 (C)



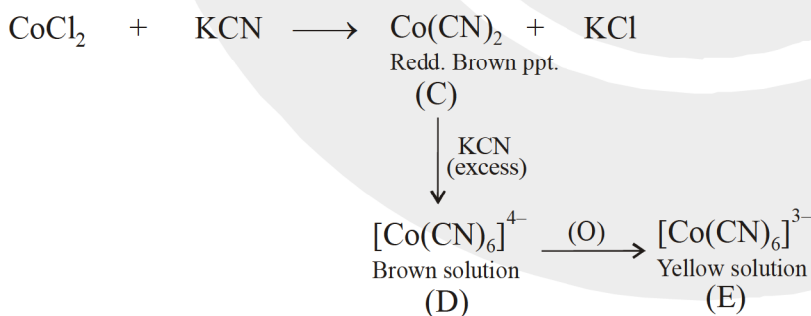
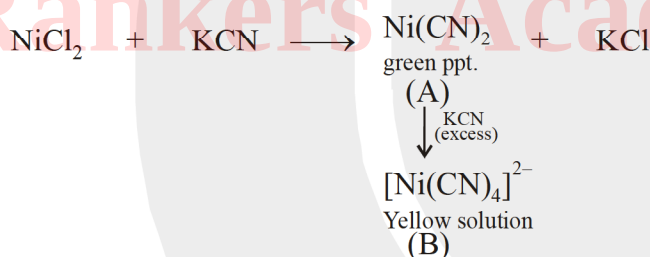
Q.2 (A)



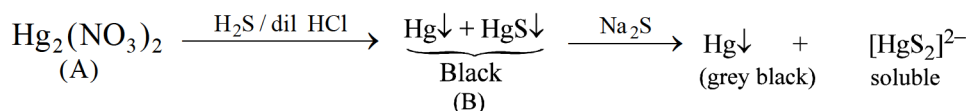
Q.3 (C)



Q.4 (A)

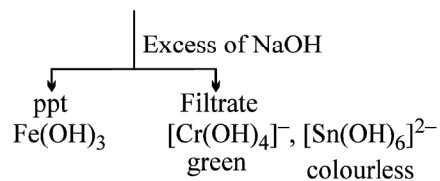
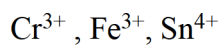


Q.5 (D)

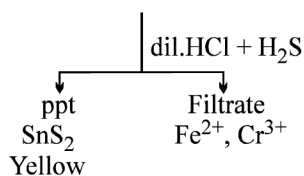


Q.6 (B)

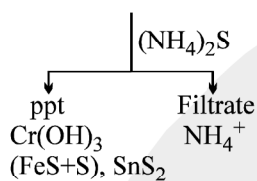
(B)



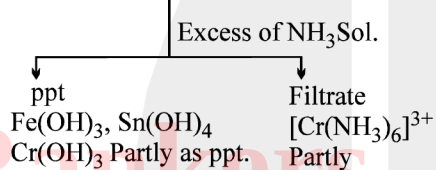
(A) $\text{Cr}^{3+}, \text{Fe}^{3+}, \text{Sn}^{4+}$



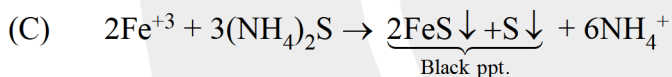
(C) $\text{Cr}^{3+}, \text{Fe}^{3+}, \text{Sn}^{4+}$



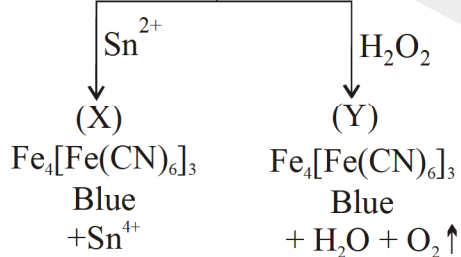
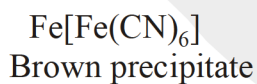
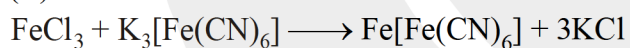
(D) $\text{Cr}^{3+}, \text{Fe}^{3+}, \text{Sn}^{4+}$



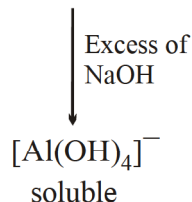
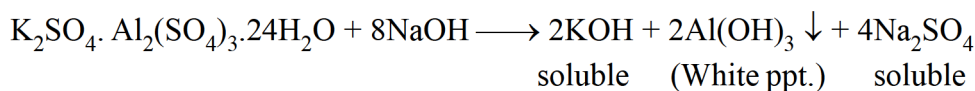
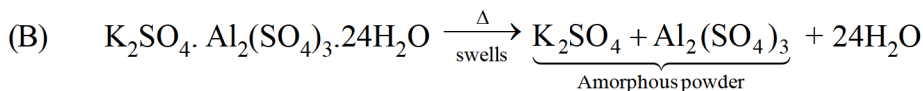
Q.7 (C)



Q.8 (A)



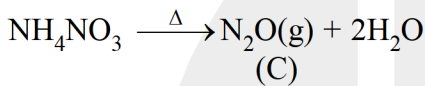
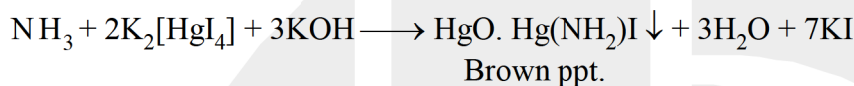
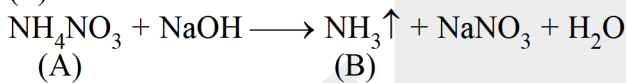
Q.9 (B)



Q.10 (B)

Q.11 (A)

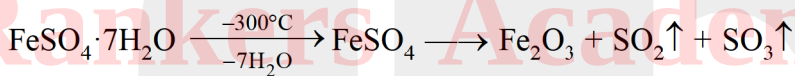
Q.12 (B)



Q.13 (D)

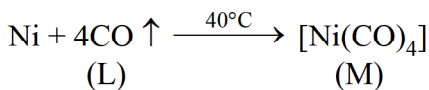
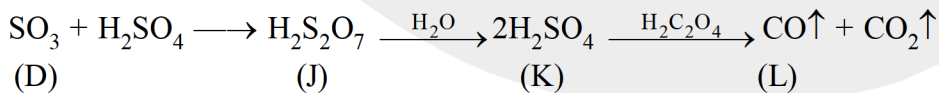
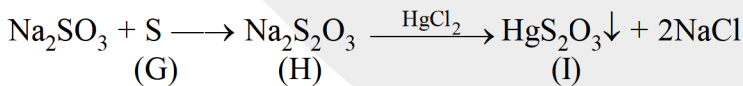
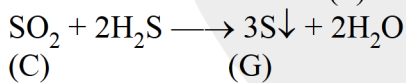
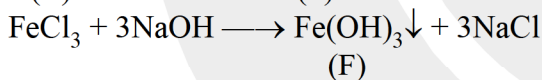
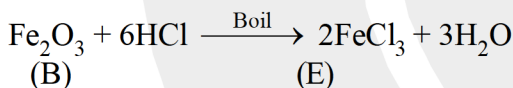
Q.14 (C)

Q.15 (B)

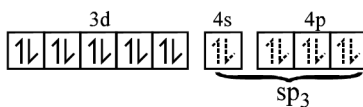


green
(A)

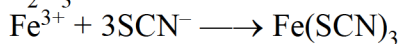
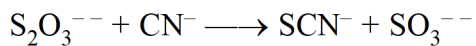
black



Ans. (i)

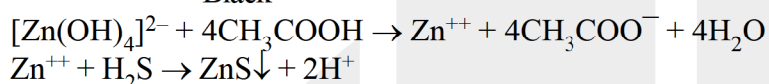
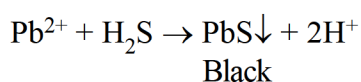
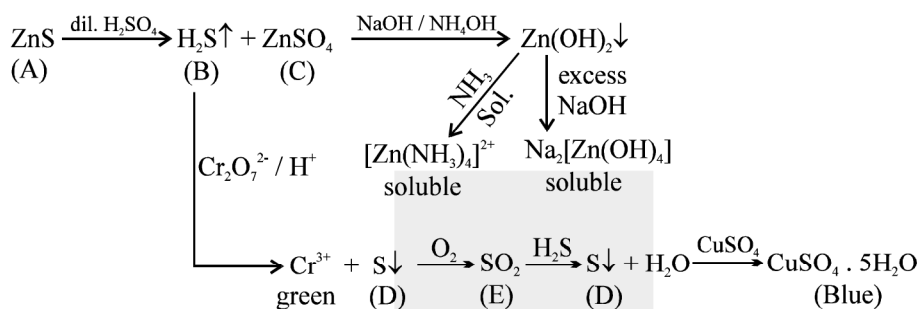
Ni in $[\text{Ni}(\text{CO})_4]$ 

Ans. (iii)



Blood red colouration

- Q.16 (C)
Q.17 (B)
Q.18 (D)

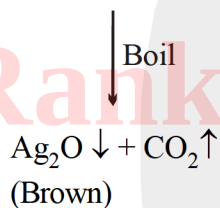


- Q.19 (C)

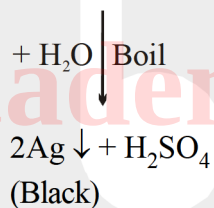
SO₂ Vapour may also be evolved from SO₃²⁻ and S₂O₃²⁻ ion by decomposition with dil. H₂SO₄

- Q.20 (C)

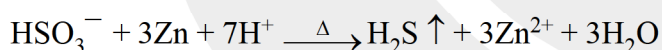
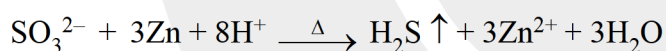
Aq. suspension of Ag₂CO₃



Aq. suspension of Ag₂SO₃



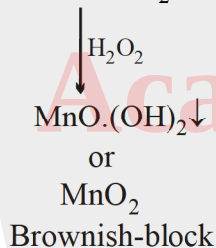
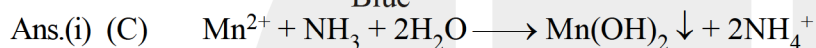
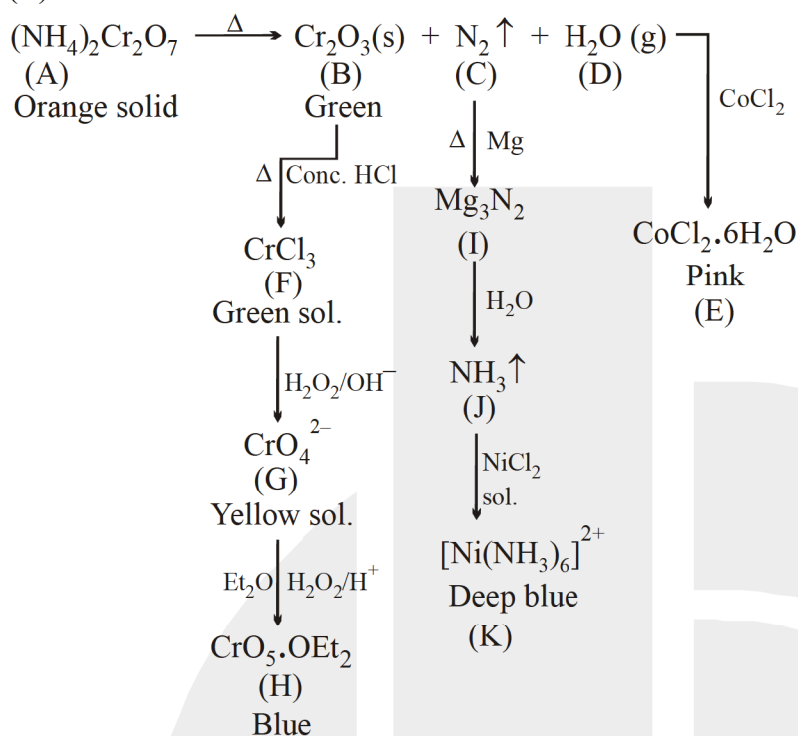
- Q.21 (B)



Q.22 (C)

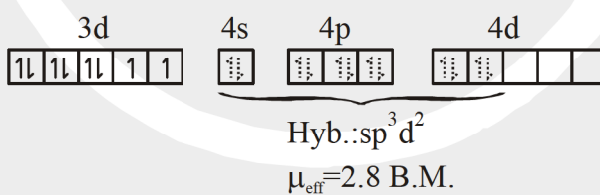
Q.23 (B)

Q.24 (D)

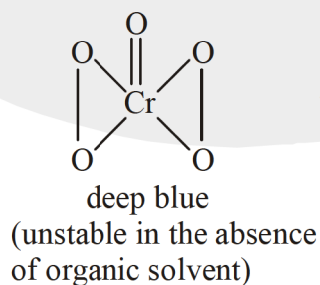


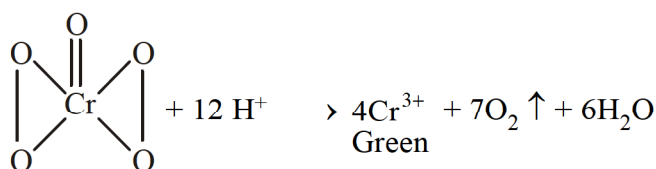
Ans.(ii) (B)

Ni^{2+} in $[\text{Ni}(\text{NH}_3)_6]^{2+}$



Ans.(iii)(D)



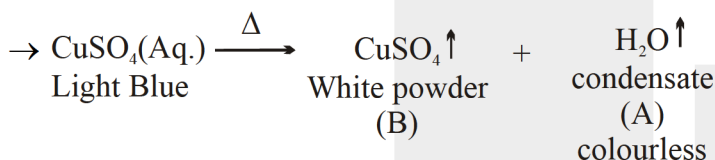


→ CrO₅ is stabilized in presence of ether, Amyl alcohol, amylacetate

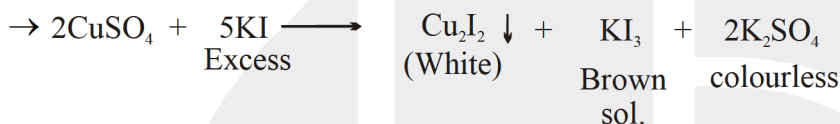
Q.25 (C)

Q.26 (C)

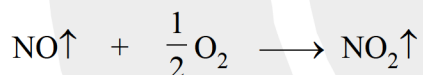
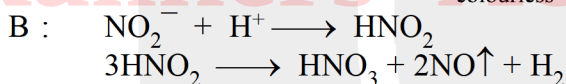
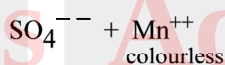
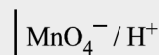
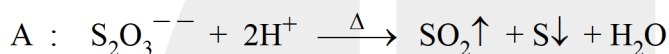
Q.27 (D)



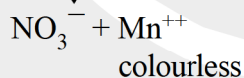
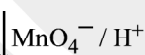
→ On adding H₂O to KI sol. it is simply diluted without any colour change.



Q.28 (AB)



colourless atm air Brown



Q.29 (ABD)

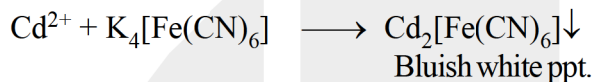
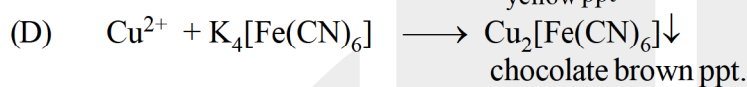
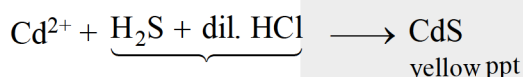
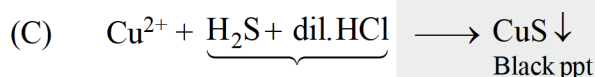
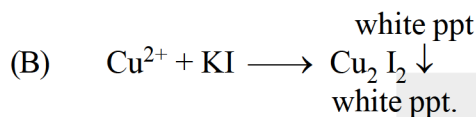
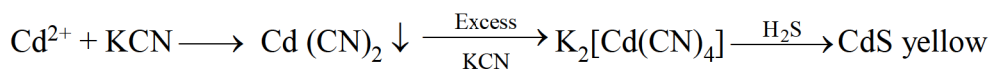
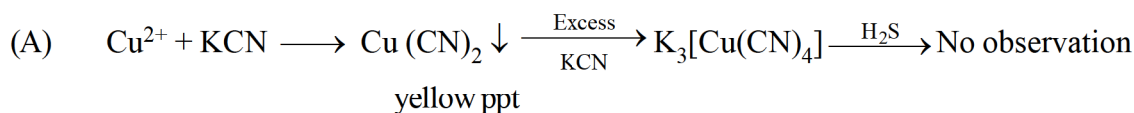
- (A) Comproportionation
- (B) Comproportionation
- (C) Dispropotionation
- (D) Comproportionation

Q.30 (ACD)

HgS is not soluble in hot and concentrated nitric acid

Metallic sulphides are soluble in hot and concentrated HNO₃ except HgS

Q.31 (ABCD)



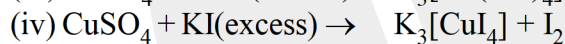
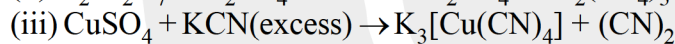
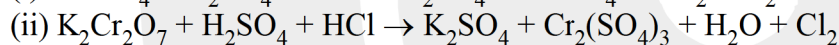
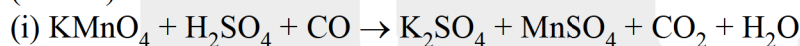
Q.32 (AC)

Q.33 (ABC)

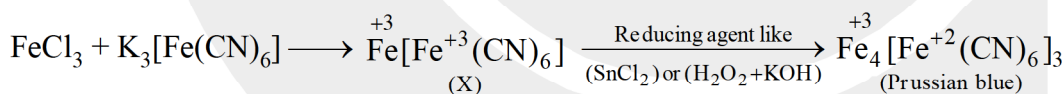
Q.34 (ABCD)

Carbonates of Mg^{2+} , Hg^{2+} , Pb^{2+} and Ag^+ are water insoluble whereas their bicarbonates are water soluble.

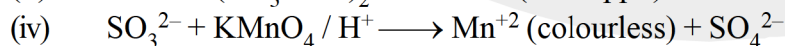
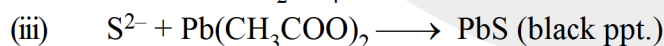
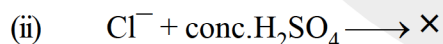
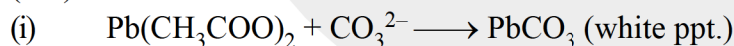
Q.35 (ABCD)



Q.36 (AB)



Q.37 (AC)



Q.38 (A) $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ (B) FeSO_4 (C) $7\text{H}_2\text{O}$ (D) $[\text{Fe}(\text{H}_2\text{O})_5 \text{NO}(\text{SO}_4)]$
(E) Fe_2O_3 (F) SO_2 (G) SO_3 (H) Cr^{3+} solution

Q.39 (A) CrCl_3 (B) $\text{Cr}(\text{OH})_3$ (C) Na_2CrO_4 (D) $\text{Na}_2\text{Cr}_2\text{O}_7$

Q.40 (A) FeBr_3 (B) AgBr (C) Br_2 (D) $\text{Fe}(\text{OH})_3$

Q.41 (A) FeSO_4 (B) BaSO_4 (C) Fe_2O_3 (D) SO_2
(E) SO_3 (F) FeCl_3 (G) S